DeployDoctor

AI-powered deployment troubleshooter

Incident diagnosis report

Generated 09 Oct 2026, 06:08 UTC

HIGH

A service or deployment is failing.

93% confidence

Affected component	Python application (SQLAlchemy/psycopg2) trying to connect to AWS RDS PostgreSQL	Category	Unknown
Analyzed	09 Oct 2026, 06:08 UTC	Model	openai/gpt-oss-120b
Analysis ID	#47	Report version	DeployDoctor 0.1.0

1 Symptom

What is failing

Application failed to start because it could not connect to the PostgreSQL RDS instance, timing out on port 5432.

2 Root cause

Why it is failing

Network connectivity to the RDS endpoint is blocked or the database server is not reachable, causing a timeout when SQLAlchemy tries to open a connection.

3 Evidence

Lines from the log that support the diagnosis

```
psycopg2.OperationalError: connection to server at
"deploydoctor-db.c9x2example.ap-south-1.rds.amazonaws.com" (10.0.12.34), port 5432 failed:
Connection timed out
```

```
sqlalchemy.exc.OperationalError: (psycopg2.OperationalError) connection to server at
"deploydoctor-db.c9x2example.ap-south-1.rds.amazonaws.com" (10.0.12.34), port 5432 failed:
Connection timed out
```

```
2026-10-05 08:21:37,498 ERROR Application startup failed. Exiting.
```

4 Fix

What to do about it

1. Verify the RDS instance is running and its endpoint is correct.
2. Check the security group attached to the RDS instance allows inbound traffic on port 5432 from the application's subnet or security group.
3. Confirm the VPC/subnet routing and network ACLs allow traffic between the app host and the RDS IP (10.0.12.34).
4. If using IAM authentication or SSL, ensure the required parameters are set correctly.
5. Restart the application after network connectivity is confirmed.

Commands

Recommendations only. Review each command before running it. DeployDoctor never executes anything.

```
aws rds describe-db-instances --db-instance-identifier deploydoctor-db --query
'DBInstances[0].Endpoint.Address'
```

```
aws ec2 describe-security-groups --group-ids <APP_SG_ID> --query
'SecurityGroups[0].IpPermissions'
```

```
nc -zv deploydoctor-db.c9x2example.ap-south-1.rds.amazonaws.com 5432
```

```
curl -s https://checkip.amazonaws.com
```

5 Prevention How to stop it happening again

- Add health?check scripts that verify DB connectivity before full startup.
- Use Terraform/CloudFormation to enforce security?group rules that allow the app subnet to reach the DB on 5432.
- Enable RDS alerts for status changes and configure retries with exponential back?off in the app.
- Document and version control network configuration changes.

DevOps Insight

The failure is a classic network?reachability issue between a service and its database, often caused by mis?configured security groups or VPC routing. Proper infrastructure?as?code practices and automated connectivity checks catch such problems before deployment.

This report was generated by an AI model and can be wrong. Review every fix before applying it. Commands are suggestions and were not executed.